Angawatch: Kenya's First Cognitive Community

Yatta Plateau, Machakos County · Power the Community 2026

1.17°S, 37.45°E · Submission Deadline: April 15, 2026 · energymentors.org/participate

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ANGAWATCH ECO-COMMUNITY

Meet Wanjiku — she built Kenya's healthcare system

[INSERT IMAGE: Kenyan nurse at community health centre — rural setting]

KES 84,230 net

- Registered Nurse, 8 years experience, Yatta sub-county
- KES 110,000 gross combined with teacher husband Kamau
- Net income KES 84,230 after PAYE, NHIF, NSSF
- Two school-age children · elderly parent visits monthly

Her basics cost KES 36,100 before food, school, or savings

KES 36,100/month

Rent (3-room Matuu): KES 18,000 — nothing owned after 25 years

Electricity (KPLC prepaid): KES 7,200 — 15–25 outages/year

Commute (boda + matatu): KES 6,600 — 2+ hours daily each way

Cooking (retail LPG): KES 1,560 — KES 1,300–1,500 per cylinder

[Visual / Chart]

Full off-grid costs 4× more — Wanjiku cannot pay that

[INSERT IMAGE: Cost comparison infographic — Off-Grid vs Grid-Interactive architecture]

+KES 1.5B

- True off-grid for 2,000 families: needs 14,400 kWh BESS (4× our design)
- Additional cost: +KES 1,500M above grid-interactive design
- Adds ~KES 8,300 to every family's monthly mortgage — unaffordable
- Engineering purity that prices out the people it claims to serve

Can a nurse afford an energy system designed for her?

KES 9/kWh

- Not designed for a utility engineer — designed for Wanjiku
- Electricity bill: KES 7,200 → KES 930/month (87% reduction)
- Reliability: 75% uptime → 99.6% (BESS + genset + KPLC backup)
- Answer: Yes — if we build houses around solar, not the reverse

We don't retrofit solar onto houses — we build houses around solar

35% cheaper

- CSEB from Yatta volcanic laterite → 35% below concrete block cost
- Passive cooling eliminates air conditioning → zero cooling load
- Roof orientation optimised for solar before architectural drawings
- Community scale unlocks institutional pricing for all components

Three decisions deliver 80% of all resident benefit

80%

[chart_07_bcr_comparison.html — Three decisions BCR comparison]

- Decision 1: Biogas + LPG SACCO → PM2.5 eliminated, KES 1,200–2,700 cooking saving (BCR 2.8)
- Decision 2: CSEB construction → 35% cost saving, zero cooling load, zero AC bill forever
- Decision 3: M-Pesa SmartMeter KES 9/kWh → 64% electricity saving (BCR 2,304)

What Angawatch Is — And What It Deliberately Is Not

- IS: A grid-interactive microgrid with islanding capability
- IS: An affordability-first community for KES 84,230-net families
- IS NOT: An off-grid vanity project — Wanjiku cannot pay 4× the BESS
- IS NOT: A retrofit — every wire laid before the first brick is placed

Yatta Plateau: 102 hectares chosen with precision

60km from Nairobi

1.17°S, 37.45°E · near Matuu, Yatta Sub-County, Machakos County

102 hectares (252 acres) at 1,450–1,550m above sea level

60km from Nairobi via A109; SGR Athi River station 25km

Existing 33kV substation Matuu 4km — grid connection KES 36M

[Visual / Chart]

Six site advantages — no compromises made

5.18 kWh/m²/day

Solar: 5.18 kWh/m²/day average (PVGIS v5.2 verified) — best within 80km of Nairobi

Elevation 1,450–1,550m ASL — passive cooling, 17–27°C year-round, zero AC

Yatta Canal 2km — gravity irrigation (unsuitable for power — too weak)

Basaltic laterite: CSEB-grade, 5% cement binder, KS 02-1070 compliant

33kV Matuu substation 4km — cheapest grid connection in region

A109 highway — direct link to Nairobi and Machakos town

[Visual / Chart]

Eight zones — every hectare of 102ha is purposeful

102 ha

Residential: 37ha (36%) — 2,000 units, hybrid tenure model

YattaFarm + chrysanthemum: 20ha (22%) — food cooperative + export flowers

Community Hub (CHC, schools, worship, recreation): 12ha (13%)

Jua Kali industrial zone: 8ha (9%) — workshops, CSEB factory, EV depot

Energy infrastructure: 3ha — 4 energy hubs, biogas plant, stormwater pond

Roads, green space, commercial strip: remaining 22ha

[Visual / Chart]

Three archetypes — every income bracket deliberately served

KES
38K–160K/mo

Wanjiku + Kamau (Nurse + Teacher): KES 110K gross → 2BR/3BR KMRC mortgage

Hassan (Boda Boda operator): KES 38K net → 1BR cooperative rental, EV boda savings

Deputy Principal: KES 160K combined → 4BR, DTI 41.2%, no AHP subsidy needed

All three archetypes drawn from KNBS 2019 Census + NCK + TSC salary data

[Visual / Chart]

2,000 units — this exact mix for one exact reason

1,130 × 2BR

150 × 1BR (42m²): cooperative rental KES 10,000–12,000/month — Hassan's entry point

1,130 × 2BR (75m²): KMRC mortgage KES 33,015/month — sweet spot for KES 110K household

700 × 3BR (93m²): KMRC mortgage KES 40,803/month — Wanjiku's family home

20 × 4BR (115m²): KES 48,615/month — Deputy Principal professional cohort

[Visual / Chart]

A KMRC mortgage builds wealth — renting builds nothing

25 years

KMRC 8.99% p.a., 25-year term — longest affordable finance in Kenya

2BR at KES 33,015/month builds full title deed after 300 payments

Renting KES 18,000/month × 25 years = KES 5.4M paid, zero equity

Yatta Plateau land appreciates as A109 corridor infrastructure improves

[Visual / Chart]

Angawatch saves each family KES 37,993 per month vs Nairobi

KES 37,993 saved

[chart_04_household_cost.html — Grouped bar: Angawatch vs Nairobi by cost category]

- Angawatch total (3BR family): KES 48,107/month including EMI + all utilities
- Equivalent Nairobi satellite: KES 86,100/month — renting, not owning
- Annual saving: KES 455,916 per year compounding toward financial security

Twenty Ekeja clusters — neighbourhood design rooted in Maa tradition

20 clusters

20 Ekeja clusters × 64–100 units = 2,000 homes in walkable neighbourhoods

Each cluster: shared courtyard, borehole access, laundry court, food forest

Social design: neighbours know neighbours — crime prevention by design

Courtyard = energy hub + EV carport + communal solar carport canopy

[Visual / Chart]

Yatta laterite makes better blocks than concrete — for 35% less

35% cheaper

Yatta volcanic laterite: only 5% cement binder (vs 8–10% standard CSEB)

KS 02-1070 certified — full Kenya Standards compliance

Build cost KES 46,500/m² VAT-inclusive vs ~KES 60,000/m² conventional

Community labour during construction: local jobs + skills transfer

[Visual / Chart]

From approval to full community in four years

2028 first occupancy

2026–27: Planning approvals, land allocation, DFI financial close

Mid-2028: First occupancy — Phase 1 (500 units, energy systems online)

2029: Phase 2 complete — 1,500 units, YattaFarm fully operational

2030: Full community — all 2,000 units, all systems active, all revenue lines running

[Visual / Chart]

5,734 MWh generated · 4,355 consumed · 1,379 exported

1,379 MWh exported

[chart_01_energy_balance.html — Stacked bar: Generation vs Demand sectors + Export]

- Solar: 5,479.85 MWh/year (SMA Sunny Design validated April 7, 2026)
- Biogas: 254 MWh/year (40-cow dairy + community waste → 773 m^3/day)
- Community demand: 4,355 MWh/year across 10 sectors (8% loss factor)
- Export: 1,379 MWh × KES 12/kWh = KES 16.5M/year net metering income

7,680 Jinko Tiger Neo bifacial panels — 3,360 kWp effective

3,360 kWp

7,680 × Jinko JKM425N-54HL4R-BDB Tiger Neo N-Type Bifacial 425W panels

Nameplate 3,264 kWp; effective 3,360 kWp with 3% bifacial rear-ground gain

Performance Ratio 0.82 — conservative Sunny Design standard

Array tilt and azimuth optimised for 1.17°S equatorial sun path

[Visual / Chart]

SMA Sunny Design Validates 5,479.85 MWh/year — April 7, 2026

- Professional simulation: Nairobi met station 71km, PVGIS v5.2 data
- PR 0.82 standard · 3,360 kWp effective DC · bifacial correction applied
- Previous hand calculation: 5,515 MWh — only 0.6% difference. Design confirmed.
- Full Sunny Design report included as Appendix B of submission document

3,600 kWh CATL LFP battery — 99.6% community uptime

99.6% uptime

18 × CATL EnerOne 200kWh LFP modules = 3,600 kWh installed capacity

3,240 kWh usable at 90% DoD — exceeds 3,027 kWh night requirement by 7%

Cummins C220N5 genset (biogas-fuelled) provides additional backup layer

Year 17 full replacement — fully costed in financial model (LFP cycle life)

[Visual / Chart]

773 m^3/day biogas — from 40-cow dairy plus community waste

254 MWh/year

40-cow commercial dairy: largest single biogas input source on site

Community organic waste provides additional continuous feedstock

Total usable biogas: 773 m^3/day (234 to cooking, 539 to genset)

539 m^3/day to Cummins C220N5 genset = 254 MWh/year electricity

[Visual / Chart]

Biogas to schools and clinics — not homes. The honest decision.

3,200 m^3/day needed

Cooking for 2,000 families needs 3,200 m^3/day — we produce only 773 m^3/day

Spreading thin biogas to 2,000 homes achieves nothing meaningful for anyone

CHC + 2 schools + market kitchen: biogas eliminates PM2.5 for 400 children daily

Homes use SACCO bulk LPG (Phase 1) → solar induction cooking (Phase 3, Year 10+)

[Visual / Chart]

Three-phase cooking strategy saves KES 100M in CAPEX

KES 100M saved

Phase 1: SACCO bulk LPG for all homes — immediate charcoal replacement, Day 1

Phase 2: Biogas to CHC, 2 schools, market — PM2.5 eliminated at institutions

Phase 3 (Year 10+): Solar induction for homes as generation surplus grows

Full induction from Day 1 = +KES 100M CAPEX + 3× solar capacity — unviable

[Visual / Chart]

KES 9/kWh via *384# — the Kenyan innovation no other team can replicate

KES 9/kWh

USSD code *384# on any Safaricom line — no smartphone or data connection needed

Purchase tokens in any amount, at any time — no minimum unit purchase barrier

Net metering credits automatically deposited to M-Pesa wallet monthly

KES 2.5M software development → BCR 2,304 (KES 5.76B lifetime family saving)

[Visual / Chart]

July dispatch: solar peaks at 2,600kW noon, exports 800kW to grid

2,600 kW peak

[24-hour power flow chart for July — solar, BESS, genset, demand, export by hour]

- 5am–8am: Cummins genset covers pre-dawn demand (biogas-fuelled, zero diesel)
- 8am–10am: Solar ramps up, genset shuts down at 9am automatically
- 10am–2pm: 2,600kW peak — BESS charging + 800kW exported to KPLC grid
- 2pm–8pm: Solar declining, BESS discharging — no genset required

April: solar drops 46% — the stress month we planned for

300 MWh (April)

[chart_02_monthly_solar.html — 12-month bar chart with April highlighted amber]

- April solar: 300 MWh vs July peak 510 MWh — 41% of best month
- Long rains cloud cover: irradiance drops to 2.8–3.2 kWh/m²/day
- Genset runs 14hr/day in April (vs 8hr dry season) — biogas-fuelled
- KPLC import: ~22 MWh total year · all in April/May · cost KES 550K vs KES 16.5M export income

Why connecting to KPLC is the correct engineering decision

KES 36M connection

Connection cost: KES 36M (4km to Matuu 33kV substation, one-time)

Returns: KES 16.5M/year net metering income + emergency April backup

Without grid: need 4× BESS (+KES 1.5B) — economically disqualifying

Thwake Dam (50km, 20MW, 2026 operational) stabilises regional grid reliability

[Visual / Chart]

1,379 MWh exported × KES 12/kWh = KES 16.5M/year community income

KES 16.5M/year

Export tariff: KES 12/kWh — KPLC approved net metering rate

Annual export: 1,379 MWh — surplus after full community demand met

Annual income: KES 16.5M — single largest community revenue stream

Credits distributed pro-rata to residents via M-Pesa net metering wallet

[Visual / Chart]

4,355 MWh total demand — every sector first-principles calculated

4,355 MWh

[Donut/pie chart — 10 demand sectors by MWh with percentages labelled]

- Residential: 2,726 MWh (63%) · Community services: 469 MWh (11%)
- Production (Jua Kali + cold storage + farm): 509 MWh (12%)
- Infrastructure (boreholes + irrigation + EV): 366 MWh (8%)
- Distribution losses: 323 MWh (8%) — conservative vs KPLC average 15–20%

3,240 kWh usable beats 3,027 kWh required — 7% headroom plus genset

7% headroom

Night demand (9pm–6am): 3,027 kWh required (residential + security + borehole)

Installed: 3,600 kWh × 90% DoD = 3,240 kWh usable capacity

213 kWh headroom (7%) before Cummins C220N5 genset needs to start

Genset is third resilience layer — not primary, not secondary, but always available

[Visual / Chart]

Cummins C220N5 180kW genset — 25 years reliable on biogas fuel

180 kW

Cummins C220N5 natural gas genset: 180kW rated, biogas-compatible

Dry season (July): 8hr/night baseload — covers pre-dawn demand gap

April stress month: 14hr/day — compensates for 46% solar drop

Warranty: 25,000 operating hours ≈ 25 years at this duty cycle. Zero diesel.

[Visual / Chart]

$2m^2$ flat-plate per unit eliminates 1,095 MWh/year electrical demand

1,095 MWh/year

Each unit: $2m^2$ thermosiphon flat-plate solar water heater (no electricity)

Replaces 2kW electric immersion heater × 3hr/day = 6 kWh/day/unit demand

6 kWh × 2,000 units × 365 × 70% solar fraction ÷ 3 = 1,095 MWh/year saved

Thermal solar 3× more efficient per m^2 than PV for hot water — correct tech choice

[Visual / Chart]

Five e-matatus, 50 electric bodas, 2-minute battery swap station

2 minutes

5 × Roam Rapid e-matatus: community transport at KES 80–100 per trip

50 × Roam Air electric bodas: KES 50/day charging vs KES 200+/day petrol

Battery swap station: 1.85 kWh packs swapped in 2 minutes (Ampersand model)

EV charging demand: 269 MWh/year — fully accounted in 4,355 MWh balance

[Visual / Chart]

Five numbers that cl

- 5,734 MWh generated (solar + biogas)
- KES 9/kWh community tariff
- 99.6% annual uptime (BESS + genset)
- KES 16.5M/year net metering income
- SMA Sunny Design validated April 7, 2026

- vs 4,355 MWh consumed — 1,379 exported
- vs KPLC KES 25/kWh — 64% cheaper
- vs Yatta sub-county ~75% KPLC uptime
- from 1,379 MWh export surplus
- 5,479.85 MWh/year — to 2 decimal places

The Cognitive Community — Kenya's First

- AI flood prediction · 14-node IoT sensor mesh · Swahili SMS alerts
- LSTM neural network: 85–95% accuracy · 1–6 hour advance warning
- BCR 61.5 — most cost-effective community safety system per person in Kenya
- Already built: Red Bull Basement working prototype — hardware, not concept

Kenya 2024: 291 killed · Machakos County directly hit

[INSERT IMAGE: Kenya flood map April 2024 / news headline collage]

291 killed

- April–May 2024: worst flood season in Kenya in 30 years (national records)
- 291 fatalities nationally · 165,000 people displaced across affected counties
- Machakos County: Yatta River and Athi River tributaries flooding sub-county roads
- The 2024 extreme event is in our LSTM training data — we learned from it

Yatta Plateau is NOT flood-prone — this is precautionary excellence

[INSERT IMAGE: Elevation [illegible] Yatta Plateau vs Athi River floodplain]

1,450m ASL

- Yatta Plateau: 1,450–1,550m ASL — well above Athi River floodplain
- Machakos River floods the lowlands — Angawatch sits upland, protected
- Our IoT system monitors internal drainage basin: canals, ponds, bioswales
- We guard against the 1-in-50-year stormwater event on the plateau, not river floods

We already built it — Red Bull Basement working hardware

Working prototype

Platform: ESP32 microcontrollers + LSTM neural network on Raspberry Pi

Built at Red Bull Basement programme, 2025 — functional prototype

Status: generates real Swahili SMS flood alerts from real sensor data

Hardware available for physical inspection — documented in submission Appendix C

[Visual / Chart]

14 sensor nodes, one mesh — canal upstream to community boundary

14 nodes

Nodes 1–3: Yatta Canal upstream — flow rate + water level gauges

Nodes 4–6: Retention ponds — fill level + overflow trigger sensors

Nodes 7–9: Bioswale network — flow direction + accumulation sensors

Nodes 10–14: YattaFarm soil moisture + community boundary rainfall gauges

[Visual / Chart]

LSTM: 85–95% accuracy, 1–6 hour advance flood warning

85–95% accurate

Long Short-Term Memory (LSTM) recurrent neural network — state of the art

Trained on 14 years Machakos County rainfall and hydrological flow data

Accuracy 85–95% validated via k-fold cross-validation on held-out data

Lead time: 1 hour minimum (fast events), 6 hours maximum (slow accumulation)

[Visual / Chart]

Sluice gates open automatically — 2-second response, no human required

2 seconds

Actuated sluice gates at all retention pond outlets (motorised actuators)

LSTM triggers gate-open command automatically at WARNING threshold

2-second electrical actuation — faster than any human decision sequence

Human override available at Eco-Hub command centre — not the primary path

[Visual / Chart]

Three-level Swahili alert: WATCH • WARNING • EMERGENCY

2,000 families

YELLOW WATCH: "Angalizo: mvua nyingi inatarajiwa. Hakikisha mifereji wazi."

ORANGE WARNING: "Onyo: mafuriko yanawezekana. Piga hatua za tahadhari sasa."

RED EMERGENCY: "Hatari ya dharura! Hamia eneo salama mara moja."

Three-level system prevents alert fatigue — WATCH frequent, EMERGENCY rare

[Visual / Chart]

Works when grid AND cellular both fail — offline-first architecture

0 cloud dependencies

Edge inference: LSTM runs locally on each node's Raspberry Pi 4 — no cloud

BESS-backed: all sensor nodes powered from community 3,600 kWh BESS

Acoustic sirens: BESS-powered physical alert for zero-cellular scenarios

Mesh networking: nodes communicate peer-to-peer if cellular towers fail

[Visual / Chart]

Same sensor mesh → precision irrigation → 21.3M litres/year saved

21.3M litres/year

Soil moisture sensors repurposed: flood detection AND irrigation control in software

Precision drip: irrigate only when soil moisture drops below calibrated threshold

Water saving vs schedule-based irrigation: 21.3M litres/year = KES 380K value

YattaFarm yield increase: 15–20% from optimal soil moisture — zero extra hardware

[Visual / Chart]

Sensor data as a product — KES 425K/year from four data buyers

KES 425K/year

Pula Advisors: hyperlocal data for parametric crop insurance pricing

ACRE Africa: smallholder weather index insurance trigger calibration

Machakos County Government: early warning planning and risk assessment

KALRO: soil moisture and rainfall data for agricultural research programmes

[Visual / Chart]

Eco-Hub command centre: flood + energy + farm on one screen

Dual 4K display

Dual 4K display at Eco-Hub community building, operated by cooperative committee

Left screen: live 14-node flood telemetry + LSTM prediction confidence levels

Right screen: energy production + BESS state of charge + grid export + farm data

Designed for elected community committee members — not engineers only

[Visual / Chart]

KES 1.16M protects 8,090 people — BCR 61.5

BCR 61.5

Total IoT system cost: KES 1.16M (hardware + installation + year 1 communications)

People protected: 8,090 residents (entire Angawatch community)

Cost per person protected: KES 143 — cheaper than one emergency room visit

Benefit-cost ratio 61.5: every KES 1 spent generates KES 61.50 in verified benefit

[Visual / Chart]

We don't just survive the climate. We master it.

[INSERT IMAGE: Night-time aerial illustration of Angawatch community with sensor node indicator lights]

Master it

- 14 nodes watch every raindrop before it becomes a community emergency
- LSTM predicts · actuators respond · residents sleep safely — all automatic
- Same sensors irrigate the farm, earn telemetry revenue, price insurance
- Kenya's first community that actively learns from its own environment

KES 12,213M — What It Buys Across 21 Components

- Housing construction (2,000 units): KES 6,840M — the core product
- Energy infrastructure (solar + BESS + biogas + genset): KES 2,100M
- Site infrastructure (roads, water, sanitation, grid): KES 1,800M
- Community facilities (CHC, schools, Jua Kali, Eco-Hub): KES 850M

Six sources, one bankable capital stack — KES 12,213M total

KES 12,213M

[chart_05_capital_stack.html — Donut chart with DFI dominant at 65.8%]

- DFI (AfDB/IFC/AIIB): KES 8,037M (65.8%) — construction bond, repaid by unit sales
- AHP Programme: KES 1,680M (13.8%) — non-negotiable for 3BR affordability
- County Bond: KES 1,000M (8.2%) · SACCO: KES 800M (6.5%)
- Carbon forward KES 163M + Equity KES 532M: community and impact investor contributions

DFI = construction finance · KMRC = take-out · Not the same shilling

Zero overlap

DFI KES 8,037M: construction bridge bond, drawn 2027–2030 during build phase

KMRC mortgages: 1,850 households × ~KES 3.5M avg = KES 6.5B over 25 years

KMRC proceeds retire the DFI bridge as each unit is sold — standard developer finance

AHP reduces price per unit independently — does not overlap with DFI or KMRC

[Visual / Chart]

DTI 48.4% on 3BR — AHP subsidy is the correct policy response

48.4%

3BR EMI KES 40,803 ÷ net income KES 84,230 = 48.4% debt-to-income ratio

KMRC guideline: 40% maximum for standard approval without assistance

AHP subsidy reduces EMI by KES 5,000–8,000 → brings DTI to 42–45%

700 units × ~KES 1.5M AHP = KES 1,050M — within AHP programme parameters

[Visual / Chart]

The 2BR at KES 33,015 — the sweet spot that needs no subsidy

39.2% DTI

2BR EMI KES 33,015 ÷ net income KES 84,230 = 39.2% DTI

Below KMRC 40% guideline — standard approval, no AHP subsidy required

1,130 × 2BR units: largest tier, designed specifically for KES 110K households

The product that proves the financial model without optimistic assumptions

[Visual / Chart]

KES 608,968 total cash at purchase — 3.4 years saving at KES 15K/month

3.4 years

2BR total unit cost: KES 3,937,500 (VAT-inclusive at KES 46,500/m²)

10% KMRC deposit: KES 393,750

Legal fees + KMRC application + stamp duty: KES 215,218

Saving KES 15,000/month: 3.4 years to full deposit — achievable for dual-income family

[Visual / Chart]

KES 182.5M per year — nine streams, zero single-point-of-failure

KES 182.5M/year

[chart_06_revenue_streams.html — Horizontal bar chart of 9 revenue streams by KES M/year]

- Enterprise (Jua Kali + commercial strip): KES 92M — dominant but not monopoly
- YattaFarm + chrysanthemum exports: KES 25.4M · Cooperative rentals: KES 19.8M
- Grid export (net metering): KES 16.5M · E-matatu fleet: KES 13.1M
- BESS grid services, CSEB, CHC, telemetry: KES 15.75M across four additional streams

KES 79.3M net surplus — residents own the outcome

KES 79.3M/year

Revenue KES 182.5M − OPEX KES 68M − tax KES 35.2M = KES 79.3M net surplus

40%: Resident dividend — KES 1,320 per family per month via M-Pesa

30%: BESS replacement and maintenance reserve (accumulates for Year 17)

20%: Community expansion fund | 10%: County development levy

[Visual / Chart]

Honest 25-year curve — Year 17 BESS replacement — breakeven Year 25

Year 25 breakeven

[chart_03_cashflow.html — Line chart cumulative position Y1 to Y25 with Y17 BESS dip]

- Years 1–3: deepest deficit during construction (no revenue, full draw on DFI)
- Year 5: all revenue streams active, deficit narrowing to KES 3,796M cumulative
- Year 17: BESS replacement cash dip — fully funded by 30% maintenance reserve
- Year 25: cumulative breakeven — community fully self-funding thereafter

Equity IRR 12–15% — meets AfDB threshold, includes Year 17 BESS

22–28% social IRR

Equity IRR: 12–15% over 25 years (Year 17 BESS, OPEX escalation, AHP all included)

Social IRR: 22–28% (resident health gains, time savings, generational wealth)

AfDB minimum benchmark: 12% equity IRR for infrastructure — Angawatch meets it

Carbon credits (22,910 t CO_2/year Verra VCS) not in financial IRR — additional upside

[Visual / Chart]

22,910 tonnes CO_2/year saved — Verra VCS registered

22,910 t CO_2/year

Solar displacing KPLC: 18,200 t CO_2/year (grid carbon factor 0.33 kgCO_2/kWh)

Biogas replacing charcoal at institutions: 2,200 t CO_2/year

EV fleet replacing petrol vehicles: 2,510 t CO_2/year

Carbon forward sale: KES 163M upfront from 10-year Verra VCS verified forward contract

[Visual / Chart]

Year 5 snapshot — all systems active, net surplus positive

All 2,000 units occupied · all 9 revenue streams fully operational

Grid export: 1,379 MWh × KES 12 = KES 16.5M (first full annual cycle)

Net surplus KES 79.3M — first resident dividend deposited via M-Pesa

BESS maintenance reserve fund begins accumulating for Year 17 replacement

[Visual / Chart]

M-Pesa 2,304% return · IoT BCR 61.5 · Biogas BCR 2.8

2,304%

[chart_07_bcr_comparison.html — Log-scale BCR comparison bar chart]

- M-Pesa SmartMeter: KES 2.5M investment → KES 5.76B lifetime saving (2,304% return)
- IoT flood system: KES 1.16M protects KES 71.3M in assets and lives (BCR 61.5)
- Biogas plant: KES 65M → KES 182M in health, cooking, and electricity benefits (BCR 2.8)
- All three decisions positive · software returns eclipse hardware by orders of magnitude

Tight affordability, AHP essential — but achievable and verified

[INSERT IMAGE: Honest risk assessment — tight but achievable with specific mitigations]

Achievable

- Affordability: 3BR at 48.4% DTI — AHP is required. We designed to qualify.
- Timeline: aggressive but achievable — Kenya's SGR was on-time and on-budget
- Numbers: every figure third-party verified (Sunny Design, KMRC calculator, KNBS)
- Risk: real but mitigated — phased build, diverse revenue, BESS maintenance reserve

Angawatch vs Yatta Sub-Cou

- Energy Cost
- Grid Uptime
- Housing
- PM2.5 (cooking)
- Carbon Footprint

- KES 9/kWh (Angawatch) vs KES 25/kWh (KPLC) — 64% cheaper
- 99.6% (Angawatch) vs ~75% (KPLC Yatta) — 24.6pp improvement
- Builds equity (Angawatch) vs pays rent forever (today)
- Near-zero (Angawatch) vs 300–1,000 µg/m^3 charcoal (today)
- 22,910 t CO_2/year saved — net positive vs grid/charcoal baseline

22,910 tonnes CO_2/year — removing 4,580 cars from Kenya's roads

4,580 cars

Solar displacing grid electricity: 18,200 t CO_2/year avoided

Biogas replacing charcoal at institutions: 2,200 t CO_2/year avoided

EV fleet replacing petrol motorcycles and matatus: 2,510 t CO_2/year avoided

Total: 22,910 t CO_2/year = equivalent to removing 4,580 passenger cars annually

[Visual / Chart]

400 children breathe clean air daily — PM2.5 eliminated from institutions

[chart_08_pm25_reduction.html — Horizontal bar with WHO safe limit reference line at 15 µg/m³]

- School kitchens (Phase 2 biogas): PM2.5 drops from 650 to 8 µg/m³ (98.8% reduction)
- Home kitchens (Phase 1 LPG): PM2.5 drops from 650 to 20 µg/m³ (immediately, Day 1)
- Community PM2.5 eliminated vs charcoal baseline: ~13.0 tonnes/year total
- WHO 24-hour safe limit: 15 µg/m³ — Angawatch Phase 2 institutions: 8 µg/m³

Cap 490 cooperative — residents vote on tariff, own the energy system

Co-op owned

Registered Cooperative Society under Cap 490 — full Kenyan cooperative law

Energy tariff voted annually by residents — any change needs 51% majority vote

All community energy assets: cooperative-owned, not private utility company

Board: 9 elected residents + 2 county nominees + 1 DFI observer (non-voting)

[Visual / Chart]

One model, 47 counties — the Angawatch template is open and transferable

47 counties

Core design: CSEB + M-Pesa SmartMeter + biogas + BESS — county-agnostic

Site selection: six-variable scorecard works for any 100ha site in Kenya

Cooperative legal structure: Cap 490 applies in all 47 counties without modification

IoT mesh: same ESP32 + LSTM — retrain only the model on local rainfall data

[Visual / Chart]

Four county types — four minimum adaptations, everything else identical

4 archetypes

Semi-arid (Turkana, Marsabit): larger BESS, more boreholes, less biogas feedstock

Highlands (Nyeri, Kirinyaga): smaller BESS (lower variance), possible micro-hydro supplement

Lake Basin (Kisumu, Homabay): fish cold-chain integration, lakefront orientation

Coast (Mombasa, Kilifi): seawater desalination instead of borehole, wind supplement

[Visual / Chart]

47 × 2,000 = 470,000 Kenyans l

- Families served
- Residents lifted
- Annual carbon saving
- KMRC mortgages created
- Solar installed nationally

- 94,000 families across all 47 counties
- ~470,000 Kenyans in clean-energy communities
- 1.08M tonnes CO_2/year at full 47-county deployment
- 94,000 — largest affordable housing programme in Kenya's history
- 157.9 MWp — equivalent to one full thermal power station

Three asks — three critical partnerships to begin

Ask 1: DFI partnership (AfDB / IFC / AIIB) — KES 8,037M construction finance

Ask 2: AHP qualification — non-negotiable for 3BR at KES 84,230 net income

Ask 3: Machakos County land allocation — 102ha near Matuu, Yatta Sub-County

All three asks are specific, actionable, and supported by existing government programmes

[Visual / Chart]

Not a concept — a verified, prototyped, documented submission

24 sections

Red Bull Basement prototype: working ESP32 + LSTM hardware — available for inspection

SMA Sunny Design report: 5,479.85 MWh/year validated April 7, 2026 (Appendix B)

24 submission sections: every claim sourced, every number independently calculated

STS token M-Pesa SmartMeter architecture: full technical specification complete

[Visual / Chart]

Angawatch. Built for Wanjiku. Powered by Kenya.

2,000 families · 8,090 residents · 102 hectares · 5,734 MWh/year

Power the Community 2026 · Energy Mentors International · April 15, 2026

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